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# Real-time blood donor prediction and campaign optimization using a DNN-autoencoder and a MERN system

*A reading brief on a proposed blood-donation platform: what it claims to fix, how the stack is split, and what the public abstract does not prove.*

Vikkiramapandian D, Ajay M, Chowmyan R, Ganesh Kumar A. Proceedings of the 3rd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS), Erode, India, 11–13 June 2025, pp. 656–661. doi:10.1109/ICSSAS66150.2025.11080959.

*This paper is a system proposal. It describes a web platform and a hybrid neural model for donor availability and campaign participation. The public abstract does not report accuracy, dataset size, or a comparison against another model.*

## Full citation

[5] D. Vikkiramapandian, M. Ajay, R. Chowmyan, and A. Ganesh Kumar, "Real-time blood donor prediction and campaign optimization using DNN-autoencoder and MERN architecture," in Proc. 3rd Int. Conf. Self Sustainable Artificial Intelligence Systems (ICSSAS), Erode, India, 11–13 Jun. 2025, pp. 656–661, doi: 10.1109/ICSSAS66150.2025.11080959.

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The reference list you photographed prints this item without author names, and the DOI string is short one "S" (ICSSA66150). The proceedings table of contents and the IEEE DOI use ICSSAS66150. Page span 656–661 follows the proceedings contents: this paper begins on page 656 and the next paper begins on page 662. The DOI record dates the publication 11 June 2025.

## Record at a glance

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|---|---|
| **Authors** | D. Vikkiramapandian; M. Ajay; R. Chowmyan; A. Ganesh Kumar. The DOI record lists Ajay as Mr Ajay. |
| **Affiliation** | Department of Information Technology, PSNA College of Engineering and Technology, Dindigul, India |
| **Venue** | 3rd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS 2025), Erode, India, 11–13 June 2025 |
| **Paper type** | Conference paper proposing a blood donation management system |
| **Software stack** | MERN: MongoDB, Express.js, React.js, and Node.js |
| **Model** | A hybrid deep neural network and autoencoder, used for donor-availability prediction and campaign participation |
| **Operators** | Donors, recipients, and camp organizers, with an organizer dashboard |
| **What is public** | The problem statement, the architecture roles, and the feature list. Numerical results are not in the abstract. |

## Problem the authors set out

The paper starts from a blood donation management system that is still largely manual. The authors list the failures that follow from that: the same donor data stored more than once, weak matching between a donor and a request, slow action when blood is needed urgently, and no reliable picture of stock or camps as they change.

They also treat coordination and trust as system failures, not only as staffing problems. Donors, recipients, and organizers do not share one current view. There is no dependable way to tell a donor that a camp is coming or that a unit is needed now. Sensitive donor data is not protected by a serious authentication design.

Those are the design requirements. Any later claim that the system “solves” blood shortage has to be read against this list: less duplicate data, faster matching, faster emergencies, a live camp calendar, notifications, and stronger access control.

## What they propose

The proposed system has two layers that do different jobs. The MERN application runs registration, matching, camp scheduling, and the screens people use. The hybrid deep neural network and autoencoder is the prediction piece. The abstract assigns that model two tasks: estimate whether a donor is available, and support decisions about who should be asked to join a campaign.

The application work the authors say the platform automates:

- Donor registration, so a person is entered once and can be found again.
- Blood-group matching between a request and a suitable donor.
- Real-time scheduling of donation camps, instead of a paper roster updated after the fact.
- An organizer dashboard for running those camps.
- Notifications to donors about upcoming camps and about urgent need.
- Multi-layer authentication around sensitive records.

## How the MERN stack is split

MERN is a common web stack. In this paper each part is given an operational reason, not only a technology name.

| | |
|---|---|
| **MongoDB** | The database. The authors choose it because donor, camp, and request data do not have to sit in one rigid table, and because a NoSQL store is easier to grow as records increase. |
| **Express.js** | The web-server framework on top of Node. It is the layer that receives requests from the screens and routes them. |
| **Node.js** | The server runtime. The abstract credits Express and Node together for fast server-side processing of registration, matching, and scheduling. |
| **React.js** | The interface. Donors and organizers are meant to work in the same application, with organizers given a dashboard and donors given camp and urgent-need notices. |

Read this split as an architecture claim. MongoDB is there for flexible, growing records. Node and Express are there so matching and scheduling are not left in a spreadsheet. React is there so the two human roles, donor and organizer, can act in one place. None of that, by itself, is a machine-learning result.

## What the DNN-autoencoder is for

A deep neural network is a stack of learned layers used here, according to the abstract, as a predictor. An autoencoder is a network trained to compress a record and rebuild it. That compression is often used to strip noise or repeated structure before a second model makes a decision. The paper calls its version a hybrid: the autoencoder and the deep network are used together, not as two unrelated experiments.

The only jobs the abstract assigns to this hybrid are donor-availability prediction and campaign-participation optimization. In plain terms, the model is meant to help the service decide who is likely to be able to give, and to aim a campaign at people more likely to take part. The abstract says this raises prediction accuracy and improves the efficiency of the donation system. It does not show the number.

*Do not cite an accuracy, an F1 score, a dataset size, or a named baseline for this paper unless you have read those figures in the full IEEE PDF. They are not in the published abstract.*

## What you can and cannot claim

Claims that the public record supports:

- The authors identify redundancy, slow matching, weak emergency response, missing live updates, missing notifications, and weak security as the failures of a manual blood-donation system.
- They propose one MERN application that registers donors, matches blood groups, schedules camps, notifies donors, and authenticates users in more than one layer.
- They place a hybrid DNN-autoencoder inside that application for availability prediction and campaign optimization.
- The paper is a 2025 ICSSAS conference paper from the information-technology department at PSNA College of Engineering and Technology, Dindigul.

Claims that need the full paper, or that you should not invent:

- A measured accuracy, precision, recall, or runtime.
- The number of donors, camps, or records used for training.
- The layer sizes, learning rate, loss function, or how the autoencoder output is joined to the deep network.
- A head-to-head win against logistic regression, random forest, or another neural model.
- Evidence that the deployed system shortened emergency response in a real blood bank. The abstract describes the intended system.

## How to use this reference

Use this paper when the writing turns from donor behavior to a software design. It is the reference for a real-time operational platform: one database, one matching service, a camp calendar, push-style notices, and a model that ranks who to contact. It is not a geographic study of where donors already travel, and it is not a demand-forecasting study of units required next month.

A careful sentence in a literature review is: Vikkiramapandian et al. (2025) propose a MERN blood-donation system with a hybrid DNN-autoencoder for donor availability and campaign targeting, aimed at removing manual delay, duplicate records, and weak notification. Performance figures are

reported in the conference paper and should be quoted from that PDF.

If you are comparing it with area-based recruitment research, keep the jobs separate. An area study tells you which kind of site repeat donors already use. This paper tells you how a team proposes to register those donors, message them, and score their availability. The second does not replace the first.

## One-paragraph summary you can reuse

*Vikkiramapandian, Ajay, Chowmyan, and Ganesh Kumar (ICSSAS 2025) propose a blood donation management system on the MERN stack, with a hybrid deep neural network and autoencoder for donor-availability prediction and campaign optimization. The system is specified to register donors, match blood groups, schedule camps in real time, notify donors of camps and urgent need, and protect records with multi-layer authentication. The published abstract states the intended gains in accuracy and efficiency and does not include the evaluation numbers.*

*This brief is an original study note prepared from the IEEE abstract, the DOI author record, and the ICSSAS 2025 proceedings table of contents. It is not a copy of the paper. Model diagrams, equations, and result tables exist only in the conference PDF and are not reconstructed here.*